

Yaser Madadi

Senior Software Developer & Database Specialist

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Professional Summary:

Senior Software Developer & Database Designer with 12+ years of experience architecting and delivering **scalable, high-performance applications** across healthcare, pharmaceutical, and manufacturing sectors. Strong expertise in **full-stack development**, **Azure-based architectures**, and **relational database design**, with hands-on experience across all phases of the software development lifecycle. Proven ability to design **robust backend systems**, optimize database performance, and integrate APIs, while collaborating effectively with cross-functional teams to deliver **reliable, maintainable solutions** that meet business objectives.

Professional Certifications:

Microsoft Certified Solution Developer, UAE, Dubai, 2007

Relational Database Design and SQL, British Columbia Institute of Technology (BCIT), Burnaby, BC, 2025

MS Azure & SQL Analytics Tools, British Columbia Institute of Technology (BCIT), Burnaby, BC, 2025 (In Progress)

Applied Data Analytics, British Columbia Institute of Technology (BCIT), Burnaby, BC, (In Progress)

Areas of Expertise:

Databases: SQL Server, Azure SQL, MongoDB; Database Design, Normalization, T-SQL, Stored Procedures, Functions, Triggers; Performance Tuning, Indexing, Partitioning, Query Optimization, Backup & Recovery.

Back-End Development: C# (.NET 2–10), Web API, REST, SOA, Business Logic, ORM (Entity Framework, LINQ, ADO.Net), JWT, Error Handling, Threading, Async Processing

Front-End Development: Angular (RxJS, Angular Material, Kendo UI), React, Blazor, ASP.NET MVC, WPF

Cloud & DevOps: Azure (Function Apps, Logic Apps, App Services, Key Vault), Git, CI/CD, Docker

Methodology & Testing: Agile, Waterfall, SDLC, OOP, SOLID, Unit Testing (xUnit, Moq, NSubstitute)

Languages / Frameworks: C#, .NET Framework, ASP.Net Web API, MVC, Angular, React, Blazor, WPF, Windows Forms, LINQ, Design Patterns

Professional Experience:

Full-Stack Software Developer & Database Architect:

TelePain Solutions, Vancouver, BC

2023–Present

A healthcare technology company providing virtual healthcare services and digital health solutions.

Scalable Relational Database Design:

Management required a scalable database for patient records, doctor profiles, appointment scheduling, and visit history, ensuring data integrity, high performance, and maintainability.

- **Queue Management:** Implemented patient queue management database to optimize scheduling, balance clinician workload and reduced patient wait times.
- **Electronic Health Record (EHR):** Built visit history tracking features to enable clinicians to access detailed treatment records and follow-up plans, improving care continuity.
- **SQL Optimization:** Optimized SQL queries and application performance, improving response times and scalability for high-volume telehealth operations.

Telehealth Applications Development:

The company required three integrated applications to manage chronic pain services—robust, secure, and user-friendly solutions for both clinicians and patients.

1. Clinic Management Web Application:

Developed a web-based application for clinics to manage doctors, patients, and treatment histories, built using Angular 20, C# .NET 9.0, Azure SQL and deployed to Azure.

2. Administration and Business Management Web Application:

Developed a web application for managing and administering the entire business, built using Angular 20, C# .NET 9.0, and Azure SQL.

3. Client Portal (Mobile Application):

Contributed to the development of a mobile application that allows clients to access their cases, appointments, payments, and related information, built using React 18, C# .NET 9, and Azure SQL.

Azure & CI/CD:

- *Azure Deployment & CI/CD Automation:* Deployed all applications to Azure using GitHub Actions, enabling automated CI/CD, consistent deployments, and improved release reliability.
- *Azure Function & Logic Apps Development:* Developed serverless backend functionalities with Azure Function Apps and automated workflows using Azure Logic Apps, improving operational efficiency and reducing manual work.
- *CI/CD & Automated Testing Support:* Supported CI/CD pipelines and automated testing using GitHub Actions, streamlining deployments to Azure App Services and improving release reliability.

More Details:

- *MongoDB Logging Integration:* Implemented MongoDB as a NoSQL logging layer to offload log data from SQL databases, improving system performance and storage management. Captured and tracked all database record activities to support auditing, debugging, and exception monitoring, ensuring full traceability and reliability across applications.
- *Backend, Frontend & Database Leadership:* Fully responsible for backend, frontend, and all database design and development for two web applications; for the mobile application, contributed fully to database design and backend development, and partially contributed to the frontend.

Tech Stack:

Angular 20, React Native, .NET 9, Azure SQL, MongoDB, Azure Function Apps, Azure Logic Apps, GitHub Actions

Professional Experience:

Full-Stack Software Developer & Database Specialist:

Cobel Darou Co., Tehran, Iran

2020-2023

Cobel Darou is a major pharmaceutical company in Iran, focused on the import, registration, marketing, and distribution of prescription drugs. It operates across diverse therapeutic areas including oncology, cardiovascular, and vaccines.

Database Performance Optimization:

Optimized database performance by analyzing queries, indexing critical tables, and restructuring schema to improve read/write efficiency and reduce latency.

- *Database Schema Redesign:* Redesigned and optimized database schemas, resulting in a 40% improvement in application performance.
- *Memory-Optimized Database Conversion:* Converted a large SQL Server table containing millions of records to a memory-optimized structure, resulting in dramatic performance improvements for read and write operations, significantly reducing query response times and system bottlenecks.
- *SQL Server Table Partitioning:* Implemented table partitioning in SQL Server for a large dataset, improving query performance, manageability, and system scalability, resulting in a 30% increase in data retrieval speed.
- *Enterprise Index Tuning:* Optimized SQL Server queries, indexes, and stored procedures, significantly improving database performance and reducing response times by up to 40%.
- *ADO.NET Data Access Implementation:* Designed and implemented an ADO.NET-based data access system to ensure faster and more reliable interactions between the application and the database, optimizing database access patterns while avoiding Entity Framework and direct stored procedure calls.
- *High-Volume Data Efficiency Improvement:* Analyzed query execution plans and applied indexing strategies, partitioning, and normalization techniques to enhance efficiency for high-volume transactional data.

Performance Monitoring System (PMS)

A complex Performance Monitoring System (PMS) at Cobel Darou to manage employee target-setting, track functional and behavioral KPIs with weighted scores, and automatically calculate performance ratings and department bonus allocations.

- *Complete Software Lifecycle Implementation:* Led and Developed a Performance Monitoring System (PMS) to track employee performance, calculate rankings, and support bonus allocation decisions.
- *System Scalability Enhancement:* Contributed to end-to-end performance improvements across the PMS application, enabling smoother user operations and enhanced system scalability.
- *End-to-End Software Development:* Led the full SDLC, from requirements gathering to deployment, successfully delivering the entire project including database design, backend, and frontend development.

CI/CD Test Integration

- *Unit & Integration Testing:* Coordinated with development and QA teams to integrate unit and integration tests into the CI/CD workflow, ensuring high-quality releases.

DevOps, Deployment & Security

- *Continuous Deployment to Azure:* Implemented automated deployment pipelines using GitHub Actions to publish applications to Azure App Services after every commit, ensuring code is deployed only after passing defined test plans.

- *Azure Key Vault Integration:* Managed security and credentials securely using Azure Key Vault, maintaining compliance and protecting sensitive data.
- *Operational Efficiency Improvement:* Improved system maintainability and accountability, reducing debugging and incident resolution time by up to 30%.

Centralized Logging & Audit System:

Developed a centralized, ACID-compliant logging system in SQL Server using stored procedures, logging all inserted, updated, and deleted records within the same transaction as the original data changes.

- *Centralized Transactional Logging System:* Persisted every change as a JSON value, recording complete details of modified records, user actions, and timestamps to support auditing, tracking, and integration with other systems.
- *Transactional-Safe Logging Implementation:* Implemented transactional logging to write log entries alongside the main transaction, preventing data inconsistencies and ensuring reliable audit trails.
- *Service Broker-Based Logging Integration:* Designed a Service Broker-based mechanism to transfer logging data from SQL Server to a NoSQL database, enabling scalable storage and analysis.

Legacy Angular Applications Modernization:

Modernized existing legacy Angular applications to improve performance, maintainability, and user experience.

- *ASP.NET MVC to Angular Modernization:* Led the modernization by upgrading an ASP.NET MVC application to Angular 13, improving UI responsiveness, performance, and maintainability.
- *Frontend Refactoring & Performance Boost:* Refactored and optimized frontend components and services, reducing load times and enhancing user experience.
- *System Integration:* Ensured compatibility with .NET 7 Web API backend, maintaining seamless integration and functionality across the full-stack system.
- *OOP-Driven Refactoring:* Applied OOP principles and design patterns to refactor legacy Angular applications, improving code maintainability and reusability.

Tech Stack:

Sql Server 2022, Azure SQL, Power Bi, DataBricks, .Net 7, ASP.Net Web API, Angular 13, MongoDB, Git & GitHub Actions

Professional Experience:

Software Developer & Database Designer:

Saravel Co., Tehran, Iran

2011-2020

A company specializing in the design, manufacturing, and installation of HVAC systems (Heating, Ventilation, and Air Conditioning) and mechanical equipment for residential, commercial, and industrial projects.

Automation System:

Developed a fully integrated production automation system that streamlined invoicing, payment, production planning, manufacturing workflows, and delivery, improving operational efficiency across the company.

- *Enterprise Solution Architecture & SDLC Leadership:* Collaborated with stakeholders to gather and analyze business requirements, translating them into scalable technical solutions aligned with organizational goals. Participated in the full SDLC, including requirement analysis, system design, development, testing, deployment, and maintenance, ensuring high-quality and timely deliveries.
- *HVAC Production Workflow Management System Design:* Led the design and development of an end-to-end HVAC production system, covering all stages from customer ordering, pre-invoice generation, contract management, payment processing, production line tracking, to final delivery.
- *Database Design & SQL Performance Tuning:* Designed and optimized relational databases, creating complex stored procedures, triggers, and functions, ensuring data integrity, high performance, and maintainability.
- *Service-Oriented Architecture (SOA) Design & Integration:* Built and maintained service-oriented architecture (SOA) solutions, enabling seamless integration of internal systems and external services for scalable enterprise applications.
- *Third-Party System Connectivity via APIs:* Developed RESTful APIs to expose services and integrate third-party systems, improving interoperability and data exchange.
- *OOP & Design Patterns for Scalable & Maintainable Code:* Applied OOP principles and design patterns to develop reusable, maintainable, and scalable code, reducing technical debt and supporting future system enhancements.
- *System Activity Tracking & Reliability:* Implemented logging and error-handling mechanisms to track system activity, ensuring traceability, easier debugging, and minimized downtime.
- *Database & Application Performance Enhancement:* Improved system performance by up to 50% through database optimization, including query tuning, indexing, and schema enhancements.
- *Legacy Application Modernization:* Modernized legacy MVC and Angular applications through refactoring, resulting in improved performance and scalability.

Tech Stack:

Sql Server 2019, Power Bi, .Net Core 3.2, ASP.Net Web API, Angular 7.0, Git

Education:**Bachelor of Science Degree in Mathematics:**

Shahrood University of Technology, Shahroud, Iran

1997-2002